

Empirical Proof Record: VALIDATED

2026-05-17T19:11:51.204089+00:00

Empirical Proof Record — VALIDATED

Proof ID: 62805255794292be163b887feb6218eb...

Created: 2026-05-17T19:11:51.204089+00:00

1. Claim

For walker_m4 events in matched SEC v2 + Literary Sinew trajectories, Kimera's native Tonus primitive (Minor Component Analysis on event-conditioned delta vectors) discovers a conservation satisfying (a) linear conservation_strength > 0.5 per corpus, (b) cross-corpus discovered-coefficient cosine similarity > 0.85, and (c) polynomial-degree=2 conservation_strength > 0.95 (essentially exact nonlinear conservation). All three sub-conditions must validate.

- **Operationalisation:** Apply native MCA (smallest-eigenvalue eigendecomposition of the event-delta Gram matrix) to matched (state_before, state_after) pairs at walker_m4 events on each corpus. Compare the discovered coefficient vectors via absolute cosine similarity (sign-invariant). Repeat with polynomial degree=2 basis expansion. Headline metric = 1 if all 3 sub-conditions satisfy; 0 otherwise.
- **Threshold:** sub_conditions_satisfied >= 1.0 all_or_nothing
- **H0:** At least one sub-condition fails: either Tonus discovers a weak conservation, the discovered direction is content-dependent, or polynomial extension doesn't tighten.
- **H1:** All three sub-conditions satisfy: substrate has a content-invariant linear conservation that becomes essentially exact at polynomial degree 2.

2. Verdict

VALIDATED — observed 1.0

Tonus discovery on matched SEC v2 + Literary trajectories: sub-A (per-corpus linear strength > 0.5) = True; sub-B (cross-corpus cosine > 0.85) = True (observed 0.9784); sub-C (nonlinear strength > 0.95) = True.

3. Evidence

Pillar	Statistic	Value	95% CI	Library
tonus_conservation_discovery	sub_conditions_satisfied	1.0	—	ophamin.measuring.

4. Pre-registration

- Registered at: 2026-05-17T19:11:51.007797+00:00
- Config hash: b2ea44f75f9b468fd0f0b0e3642cf565...
- Data hash: b8340226621e18833ecf74c920dab0d0...

5. Signature

dc284af46f1fc528a19402b090173cf2f29d820ef0c679d53c961165baceafa7